Data and Applications

Project Phase 3

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Introduction

This document showcases a relational model for the mini-world we described as part of Phase 1 of this project. It is derived from the ER Diagram described in Phase 2 of this project, and subsequently converted to the normal form.

Conversion to the Relational Model

The relational model is shown in Figure 1 (located in the last page of this document).

Steps undertaken for mapping the ER Diagram to a Relational Model:

1. Each strong entity is laid out as a relation along with its simple attributes, relationships will be accounted for in the subsequent steps.
2. Sub-class hierarchy is shown by linking the super-class with its corresponding sub-class using a relation on the primary keys. The sub-class will only have its non-derived attributes represented in the diagram, it is assumed to have derived the other attributes from the super-class.
3. A primary key is chosen for each of the entities.
4. Multi-valued attributes if present, are represented as a separate relation with a foreign key to the parent entity.
4. Each weak entity is represented as a relation with all its simple attributes. Additionally, foreign keys of the identifying strong owner entities are incorporated.
5. The primary key of a weak entity is chosen as a combination of the foreign keys of the identifying owner entities and its partial key, as mandated.
6. All relationships have been incorporated using *the foreign key approach*. In essence, foreign keys of the participating entities are present as attributes of another relation.

Changes made:

1. Entities with two or more key attributes (`Report`, `Ward`) now posses a single primary key, the decision has been made either logically or by considering the length of the keys.
2. Each relation has been translated to appropriate foreign keys on entities of significance.
3. For 1:1 binary relationships (`Has`, `UsedIn`, `InhabitedBy`), we must externally enforce uniqueness of the foreign keys. Instead of introducing redundancy by adding foreign keys on both participating entities, we note this strict requirement here due to lack of means of representation.
4. Derived attributes are generally omitted from relational models as they are not relevant computationally relevant but not representationally. Thus, the derived attribute `age` has been omitted from the entities `Human` and `Ghoul`.

Conversion to the First Normal Form

In reference to the book (Elmasri & Navanthe), a relational model is set to be in 1NF if the model does not contain:

1. relations within relations
2. relations as attribute values within tuples

The First Normal Form is the relational model modified to have *no multivalued attributes, composite attributes or nested relations*. This was already taken care of in the mapping of the ER diagram to the relational model, however, we reiterate the same in more detail here.

It states that the domains of attributes must include only *atomic* values. In our relational model (Figure 1), the `Rating` relation was formerly a multi-valued attribute of the `Ghoul` relation. In the mapping of the Entity-Relationship Diagram to the Relational Model, a new table was created to handle the multivalued attribute `Rating`.

So we replaced the multivalued `rating` field in `Ghoul` with a new relation `Rating` with fields `ghoul_id` which is a foreign key to `Ghoul`'s `id`, and the atomic `rating` value, forming a one-to-many relation, and eliminating multivalued attributes from our model.

We also had a composite attribute {`division`: `number`, `city`} of the `Ward` entity in our ER model. As the mapping stated, the simple components of the composite attribute were laid down. This lead to a flat structure that was compliant with both the Mapping of ER to relational model as well as the 1NF requirements.

Both these steps were performed in the initial ER diagram to relational mapping conversion step, as prescribed by the steps provided in Chapter 9 of Elmasri and Navanthe. As a result, our initial relational model is already in 1NF, and thus, this **normalisation is not needed**.

Figure 1 shows the first normal form of our model. It is the same as the original relational model.

Conversion to the Second Normal Form

In reference to the book (Elmasri & Navanthe), a relational model is set to be in 2NF if:

- The model is in 1NF
- Every non-prime attribute is fully functionally dependent on the primary key(s) of the relation.

Additionally, it has been stated that the 2NF test need not be applied if the primary key contains only a single attribute. Our relational model contains the following relations with primary keys having multiple attributes:

- `GhoulMurder` has PK: {`Murderer`, `Victim`, `time`}, we have the following justification regarding this relationship:
  - Ward id is uniquely determined by Murderer, Victim and time. It is not partially dependent on the primary key.
  - Multiple murderers can be involved in murder cases in the same ward.
  - Multiple victims can be killed in the same ward.
  - Different murders can happen at different times in the same ward.

Hence, the ward can be uniquely determined by the full primary key itself.

- `Encounter` has the primary key: {`Senior_id`, `Junior_id`, `Ghoul_id`, `time`}, the following points bring out the lack of need of 2NF for this relation:
  - The non-prime attributes (`death_senior`, `death_junior`, `death_ghoul`, `JuniorQuinque`, `SeniorQuinque`) can be uniquely determined by the full primary key. These attributes are not partially dependent on the primary key.

- death_senior, death_junior, death_ghoul are uniquely determined by the primary key, i.e. in order to derive them, we need to know the the information of Senior_id, Junior_id, Ghoul_id as who were involved in the encounter which caused the deaths, and the time of the encounter.
- JuniorQuinque, SeniorQuinque are also uniquely determined by the primary key as we need to know the same information given by the primary key.

Hence, the non-prime attributes are not transitively dependent on the primary key.

- OneEyedGhoul has PK: {HumanPart_id, GhoulPart_id}
  - The non-prime attributes (HumanParent_id, GhoulParent_id, TransplantDonor_id) can be uniquely determined by the full primary key as in order to derive them, we need know the full information of both the human part and the ghoul part.

Thus, as one can reason from the above relations with multiple primary keys, the model (Figure 1) is already in 2NF, and thus, **normalisation is not needed**.

Conversion to the Third Normal Form

In reference to the book (Elmasri & Navanthe), a relational model is set to be in 3NF if:
- The model is in 2NF
- No non-prime attribute is transitively dependent on the primary key.

For every relation, we have used the foreign key approach as appropriate with regards to the cardinality ratio and type of participation. Hence, for every relation R, there is no attribute that is transitively dependent on any of the primary keys.

Consider all the entities in the given diagram (Figure 1), all the non-prime attributes are uniquely identified by the entirety of the primary key and no other candidate or subset of the primary key. Hence, the Third Normal Form of our relational model is identical to the Second as in Figure 1. The model is already in 3NF, and thus, **normalisation is not needed**.

In more detail:
- QuinqueUpgrades has PK: {Quinque, Kagune}, and there exists a unique investigator for the given Primary key set as a Quinque may have multiple ingredient Kagunes and vice versa.
- GhoulMurder has PK: {Murderer, Victim, time}, and as explained in 2NF propositions, the ward is uniquely identified solely by the entirety of the primary key considering all the real-world cases in the anime.
- Killing has PK: {ghoul_id}, and we assert that it does not entail any transitive dependencies. This is because an investigator may not use his own quinque for the killing of the ghoul and the ward in which this unfortunate incident occurs is solely determined by the event details. Hence, as none of the non-prime attributes depend on each other in any way, there cannot exist a transitive dependency.
- Encounter has PK: {Senior_id, Junior_id, Ghoul_id, time}, and similar logic from the Killing relation holds with the non-prime attributes JuniorQuinque and SeniorQuinque. All non-key attributes are hence, not correlated.
- OneEyedGhoul and Chimera have PK: {HumanPart_Id, GhoulPart_Id} and {ghoul_id} respectivey. One can conceptually observe that all other non-prime attributes are completely dependent on the primary key and do not hold any significance among themselves.

The Relational Model & Normal Forms

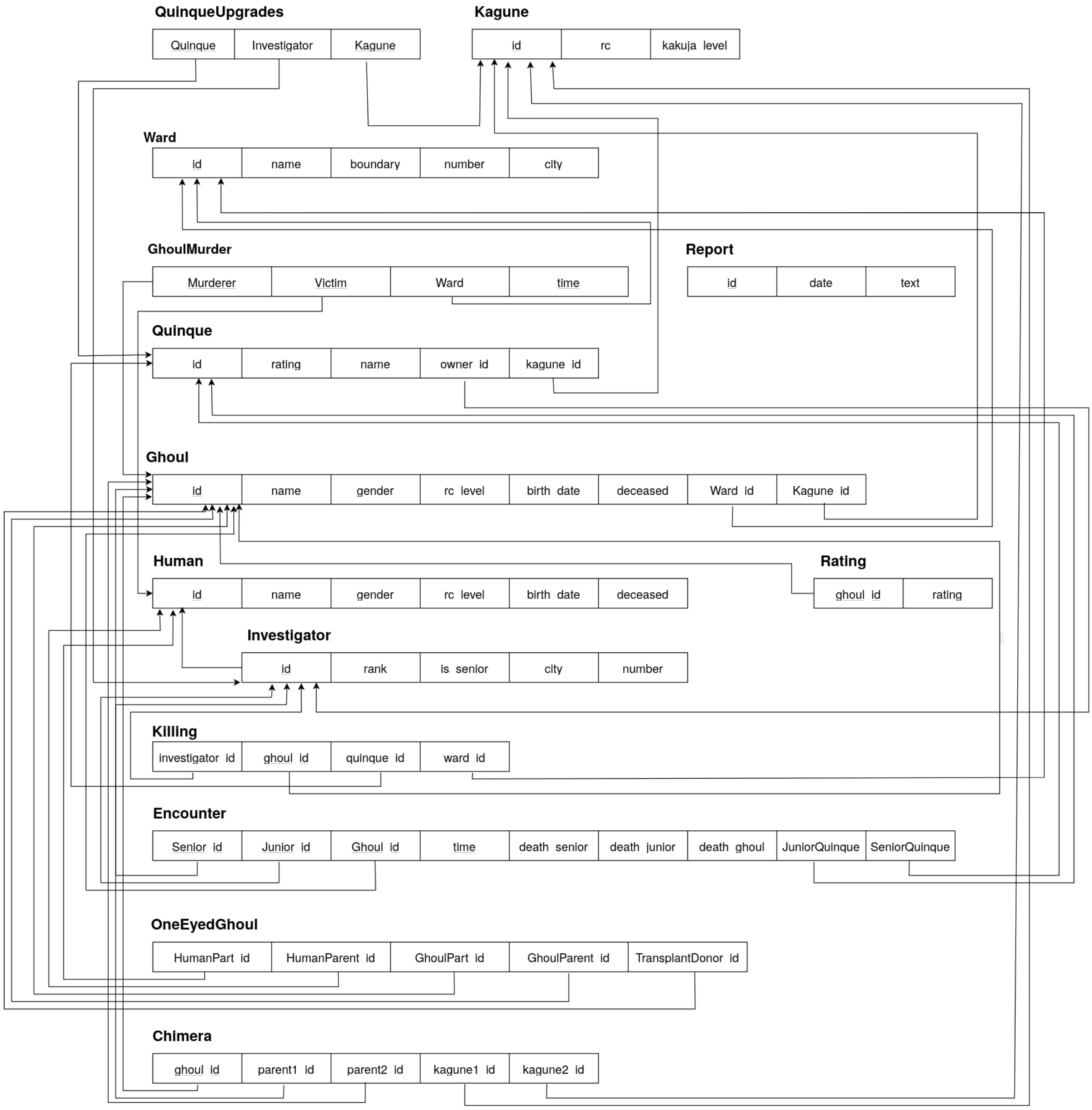

Figure 1: The Relational Model